

PROJECT PELICAN-UAV

Biomimetic Long-Endurance Aerial Platform with Variable Buoyancy Control

Conceptual Specification and Technical Pre-Design Document

Version 1.0 · Preprint — Zenodo

June 2026

Author	Hernán Diego Herrero
Affiliation	Fundación Hijos del Sol y la Basura · Córdoba, Argentina
Field	Biomimetic Aerospace Engineering · Structural Electronics · Advanced Aerostatics
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DOI	<i>Pending assignment (Zenodo)</i>

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Abstract

Background	Remote ecosystems and protected areas across Latin America and the Global South face persistent threats — illegal logging, poaching, agricultural encroachment, and illegal, unreported and unregulated (IUU) fishing — in territories where state presence is minimal and community rangers operate with scarce resources. Conventional UAV platforms (fixed-wing or rotary) are poorly suited for these contexts: they generate acoustic signatures that disturb the fauna they aim to observe, their endurance is insufficient for large-area coverage, and their cost places them beyond the reach of community conservation programs.
Objective	To propose a hybrid aerostatic-aerodynamic UAV architecture with avian morphology and flight behavior, capable of operating for extended hours without perturbing the ecosystem under observation, at a construction cost accessible to community conservation programs, and with low-observable properties that render it undetectable by the radar systems typically carried by IUU fishing vessels.
Methodology	Biomimetic conceptual design based on Pelecanidae morphology. Analysis of

	static lift vectors (H_2), multifunctional envelope design as a distributed structural battery, and closed-loop pneumatic architecture for altitude control without gas loss. No physical experiments are conducted in this version; the document constitutes a pre-design technical specification for subsequent prototyping.
Expected outcomes	An aerial platform with endurance far exceeding conventional systems of equivalent weight class, acoustic, visual and infrared signatures indistinguishable from local avifauna, radar cross-section equivalent to a large soaring bird, open-source replicable architecture, and compatibility with low-cost solar charging infrastructure built from recycled urban waste.
Keywords	Biomimetic UAV · aerostatics · hydrogen · structural battery · variable buoyancy · dynamic soaring · closed loop · long endurance · IUU fishing · low observable · ecosystem monitoring

1. General Architecture and Geometric Configuration

The platform adopts a fusiform biomimetic silhouette — analogous to a revolution volume resembling an American football viewed from the side — with a maximum geometric length close to 2.70 m and a maximum height of 0.90 m. Unlike conventional cylindrical fuselages, the exterior volume is rounded and turgid, tapering toward bow and stern, replicating the body of a real soaring bird. The architecture is built on three design pillars:

- **Hybrid, Distributed Lift:** aerostatic equilibrium via hydrogen (H_2) housed in the main envelope and in volumetric wings, combined with high-efficiency aerodynamic glide ratios (L/D).
- **Internal Functional Duct:** an internal ventral tube, not visible from the exterior, runs the length of the vehicle from bow to stern, housing critical systems and a reversible turbine that manages airflow according to flight mode.
- **Biomimetic Altitude Control:** the pelican's gular pouch (pouch) is reinterpreted as a functional volumetric expansion chamber, regulating buoyancy without venting gas to the exterior.

PELÍCANO-UAV

BOSQUEJO CONCEPTUAL APROXIMADO

TOMA DE AIRE
válvula activa
(pico)

BUCHE - CÁMARA DE
EXPANSIÓN VOLUMÉTRICA (VBC)
se expande/contrae para
control de altitud

ENVOLTURA PRINCIPAL
contiene H₂ a baja presión
+ batería estructural

ALAS VOLUMÉTRICAS
contienen H₂
+ batería estructural

COMPRESOR
ELÉCTRICO

TANQUE DE H₂
COMPRIMIDO

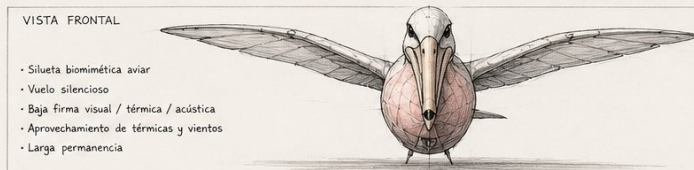
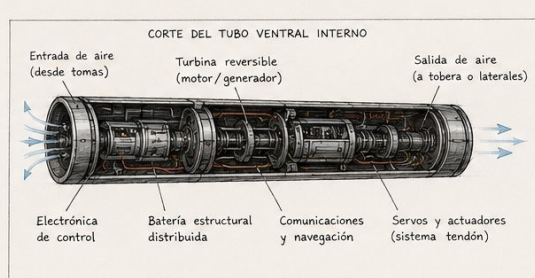
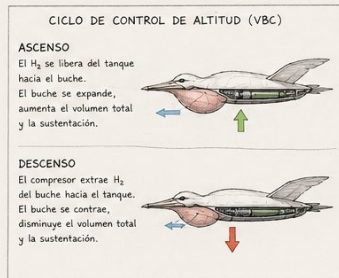
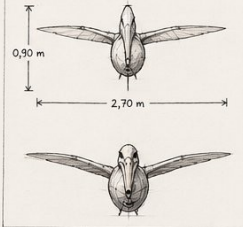
TURBINA REVERSIBLE
(motor / generador)

ELECTRÓNICA
Y BATERÍA ESTRUCTURAL
DISTRIBUIDA

TOMAS DE AIRE
LATERALES
(entradas/salidas
secundarias)

TOBERA DE COLA
(salida/entrada
de aire)

DIMENSIONES APROXIMADAS



NOTAS GENERALES

- H₂ a baja presión (~20-40 mbar)
- Sistema de ciclo cerrado (sin pérdida de gas)
- Control automático de altitud y modos
- Energía principal: atmósfera + H₂
- Energía secundaria: batería estructural
- Propulsión para maniobras y correcciones

Figure 1. Master diagram of the Pelican-UAV. General view with longitudinal cutaway (envelope, volumetric wings, pouch/VBC, compressor, H₂ tank, reversible turbine, ventral tube, lateral intakes and tail nozzle); altitude control cycle (ascent/descent); detailed cross-section of the internal ventral tube; the four flow and turbine operating modes; envelope material layer scheme; front and top views with approximate dimensions (2.70 m × 0.90 m).

2. Lift System and Altitude Control

2.1 Hydrogen as Lift Gas (H₂)

The internal volume of the main envelope and the volumetric wings is occupied by low-pressure, structurally sealed hydrogen. Using standard sea-level densities ($\rho_{\text{air}} = 1.29 \text{ kg/m}^3$; $\rho_{\text{H}_2} \approx 0.09 \text{ kg/m}^3$), the net ascensional force per unit volume is:

$$\Delta F_{\text{lift}} \approx (\rho_{\text{air}} - \rho_{\text{H}_2}) \times g \times V \approx 1.20 \text{ kg/m}^3 \times g \times V \quad (\text{Eq. 1})$$

Unlike a conventional airship, this volume is neither uniform nor singular: it is distributed across the main body, the two volumetric wings, and the ventral expansion chamber (pouch), distributing lift across the entire geometry and improving both longitudinal and transverse stability.

2.2 The Gular Pouch as Volumetric Expansion Chamber (VBC)

The central innovation of this architecture is the functional reinterpretation of the pelican's gular pouch. Far from being an aesthetic element, the pouch is the visible external manifestation of a closed-loop Variable Buoyancy Control (VBC) system:

- An electric compressor and a compressed-hydrogen tank, both housed in the vehicle's head, constitute the high-pressure gas reservoir.
 - The pouch, located ventrally immediately behind the head, acts as a low-pressure expansion chamber connected to the tank via transfer valves.
1. Ascent: hydrogen stored in the head tank is released into the pouch, which expands downward under gravity. The total displaced volume of the vehicle increases, raising net lift.
 2. Descent: the electric compressor draws hydrogen from the pouch back into the tank. The pouch contracts, total volume decreases, and net lift is reduced, generating a controlled descent vector.

This system preserves the total H_2 mass on board throughout each cycle — it is thermodynamically closed with respect to the lift gas — and requires no venting or replenishment of gas at any point during the mission.

2.3 Mass Placement and Center of Gravity

The placement of the compressor and compressed-hydrogen tank in the head is not arbitrary: it concentrates the system's densest mass at the forward end, replicating the mass distribution of a real pelican (heavy head and bill relative to the body). This produces two beneficial effects:

- It establishes a forward center of gravity that favors the natural pitch-down attitude characteristic of Pelecanidae soaring flight, without requiring additional active control surfaces for that function.
- The VBC cycle dynamically shifts the center of gravity: with the pouch full (ascent), weight shifts slightly toward the belly; with the pouch empty (descent), relative weight concentrates toward the head. This passive CG shift contributes to attitude control at no additional energy cost.

3. Internal Ventral Tube and Propulsion/Generation System

3.1 Duct Architecture

An internal duct — the ventral tube — runs the length of the vehicle, concealed beneath the outer envelope and not visible from the exterior. It houses the structural battery, control electronics, communications, servomechanisms, wiring, and, at its central section, a reversible turbine coupled to a permanent-magnet electric motor.

Airflow through this duct is actively managed via three valves: the bill (forward intake, an active servo-driven valve), the lateral intakes (secondary inlets/outlets located in laminar-flow zones of the profile), and the tail nozzle (rear outlet/inlet). The central turbine can rotate in either direction, functioning alternately as a propulsion motor or as an energy-recovery generator.

3.2 Flow System Operating Modes

The combination of variable valves and reversible turbine defines four discrete operating modes, automatically selected by the flight software according to mission phase and wind conditions:

Mode	Configuration	Function
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1 — Active propulsion	Bill closed · lateral intakes open · tail = nozzle	Lateral intakes capture air in laminar flow without generating frontal drag. The turbine, driven by the electric motor, expels flow through the tail, generating thrust. Active cruise and maneuvering mode.
2 — Forward generation	Bill open · tail closed or partial	With headwind or during active flight, incoming air enters through the bill and spins the turbine in generator mode, recharging the structural battery without H ₂ consumption or altitude loss.
3 — Rearward generation	Bill closed · tail intakes air · outlet via lateral intakes	During tailwind soaring, the rear nozzle acts as an air intake. The turbine spins in generator mode and flow is exhausted through the lateral intakes, recharging the battery during descent.
4 — Pure soaring	All valves closed · turbine at rest	Minimum aerodynamic drag configuration (maximum L/D). Lift depends exclusively on H ₂ volume; no electrical consumption or generation occurs. Default mode for maximizing endurance.

This scheme turns the propulsion system into a dual-function atmospheric energy management organ — analogous to the gill-based respiration of certain fish, where the same duct serves different purposes depending on flow direction — and drastically reduces dependence on ground-based electrical charging.

4. Multifunctional Composite Envelope

4.1 Distributed Structural Battery

The skin of the main envelope and volumetric wings acts as a laminated composite sandwich that simultaneously functions as an H₂ containment barrier and as lithium polymer electrochemical storage (structural LiPo). The electrodes and solid polymer electrolyte are designed to absorb the torsion and bending stresses inherent to flight. Design advantages include:

- Homogeneous center-of-gravity distribution across the entire structural surface, complementing the dynamic mass shift described in Section 2.3.
- Omnidirectional passive thermal dissipation optimized by the incident air boundary layer.
- Net increase in internal volume available for the ventral tube and its systems, by eliminating the discrete battery compartment.

4.2 Volumetric Wing Geometry and Tendon System

The high-aspect-ratio wings contain H₂ volume internally — they are not flat surfaces but aerostatic structures that provide chemical and aerodynamic lift simultaneously — optimized for extended gliding in low Reynolds number regimes ($Re < 5 \times 10^5$), emulating the pelican wingspan (*Pelecanus onocrotalus*, $AR \cong 9.5$).

Tail control surface deflection is achieved through a network of ultra-high molecular weight polyethylene tensile cables (UHMWPE / Dyneema, specific strength $\approx 2.4 \text{ GPa} \times \text{m}^3/\text{kg}$) coupled to stepper micro-motors and low-friction internal pulleys, replicating avian tendon biomechanics.

The result is a silent, high-positional-fidelity actuation system with electrical consumption in the milliampere range.

4.3 Containment Barrier and Permeation Mitigation

Internal to the electrochemical layer, a multilayer nanocomposite membrane (co-extruded with EVOH-type barriers or ultra-thin metallized coating) blocks the molecular diffusion of H_2 , simultaneously protecting structural battery components from environmental and chemical degradation. The target permeation rate is below the flammability threshold under normal operating conditions (H_2 LFL: 4% vol/air).

5. Atmospheric Navigation Strategy and Mode Selection

The operational philosophy of the Pelican-UAV moves away from brute-force navigation toward dynamic and thermal soaring flight, with selection among the four flow modes (Section 3.2) integrated as an additional decision variable. The flight control software integrates predictive algorithms based on onboard micro-meteorological sensors to:

3. Detect and intercept ascending thermal columns in real time, prioritizing Mode 4 (pure soaring) while the thermal remains sufficient.
4. Execute altitude transitions via the VBC system (pouch/tank) to couple with favorable wind gradients, without requiring propulsion power.
5. Switch to Mode 2 or Mode 3 (generation) when relative wind conditions permit structural battery recharge with minimal trajectory penalty — at an estimated conversion efficiency $\eta_{\text{regen}} \approx 0.35\text{--}0.45$, consistent with low tip-speed turbines.
6. Reserve Mode 1 (active propulsion) for heading corrections, station-keeping over an area of interest, and approach maneuvers.

This mode hierarchy significantly extends the lifecycle of the initial ground charge, since active propulsion — the highest-consumption mode — is reserved for a minimal fraction of the mission profile.

6. Integral Mimicry: the UAV as a Bird

The most relevant design decision in Project Pelican-UAV is not technical but methodological: an observation instrument that disturbs the system it attempts to measure introduces irreducible bias into the data. A conventional unmanned helicopter or aircraft alters fauna behavior, disperses flocks, interrupts feeding and nesting cycles. The Pelican-UAV resolves this problem at the architectural level: it does not camouflage — it is a bird.

At medium and high altitude, its silhouette, thermal soaring pattern, silence, and surface temperature are indistinguishable from those of a soaring *Pelecanus* or *Jabiru*. Animals do not react because there is nothing to detect. This is not stealth in a military sense — it is ecological coherence. And it has a direct methodological consequence: behavioral data collected from this platform are not contaminated by observer-effect bias.

7.1 Physical Signatures

- **Acoustic Signature:** absence of high-frequency noise from low-RPM propellers and UHMWPE tendon actuation. The only sound generated is below the alarm threshold of wildlife.
- **Visual Signature:** full-scale avian morphology. At operational altitudes (150–600 m), silhouette, coloration and soaring motion are indistinguishable from local avifauna.
- **Infrared Signature (IR):** distributed thermal dissipation across the entire structural surface. Without a localized hot spot (motor), the thermal profile replicates that of a passively soaring bird.
- **Radar Cross-Section (RCS):** composite polymers, continuous organic shapes and absence of dense metallic blocks reduce electromagnetic reflectivity to levels equivalent to a large soaring bird.

This convergence of signatures is not a secondary design objective — it is the primary criterion. A ranger operating the Pelican-UAV is not piloting a drone: they are releasing a bird.

7. Low-Observable Properties and IUU Fishing Vessel Detection

Illegal, unreported and unregulated (IUU) fishing represents one of the most severe threats to marine ecosystems globally, with estimated annual losses exceeding USD 23 billion [11]. Enforcement of exclusion zones and protected marine areas is chronically hampered by the radar detection capabilities of IUU vessels, which routinely identify and evade conventional monitoring aircraft and UAVs.

The Pelican-UAV presents a qualitatively different detection profile. Marine surveillance radars deployed on fishing vessels — typically X-band (8–12 GHz) or S-band (2–4 GHz) systems optimized for surface target detection — rely on metallic reflectivity thresholds that distinguish aircraft from biological targets. The platform's composite construction, organic geometry, distributed thermal signature, and soaring flight pattern place its radar return squarely within the biological clutter envelope that vessel radar operators and automated ARPA systems routinely filter out as avian noise.

8.1 Flight Pattern as Organic Camouflage

Beyond material composition, the flight behavior itself constitutes a second layer of concealment. The VBC-driven altitude transitions, the exploitation of thermal columns, the variable airspeed profile inherent to soaring, and the absence of the fixed-altitude cruise signature characteristic of powered UAVs produce a movement pattern that is algorithmically and visually consistent with a large pelagic bird — a frigate bird, a gannet, an albatross — executing normal foraging behavior over open water.

Automated vessel tracking systems trained on powered-aircraft signatures will not flag this profile. Human radar operators scanning for intruder aircraft will not distinguish it from the biological returns they habitually ignore. The organic flight pattern is not a secondary feature: it is an active low-observable mechanism emergent from the core architecture.

8.2 Operational Scenario: Coastal Protected Area

A community ranger organization operating in a marine protected area deploys the Pelican-UAV at dawn from a coastal station. The vehicle ascends silently on a coastal thermal, reaches operational altitude, and begins a systematic coverage pattern over the exclusion zone. IUU

vessels operating within the area — equipped with standard marine radar and ARPA — receive no anomalous return. The platform streams GPS-tagged imagery and AIS cross-reference data to the ground station. When a vessel operating without valid AIS transponder signal is identified, the system logs position, heading, estimated vessel class, and timestamps. The ranger organization transmits the evidence package to the relevant maritime authority.

The vessel was never aware of being observed. The evidence is admissible. The platform cost a fraction of a conventional maritime patrol aircraft.

8. Deployment Context: the Ranger and the Bird

The primary operational scenario of the Pelican-UAV is not an airfield or military base. It is the bank of a river at the edge of the Chaco forest at dawn, or the boundary of an Andean biological corridor where chainsaws were heard the previous night, or a coastal bluff overlooking a marine reserve where lights were spotted offshore at midnight.

The community ranger charges the system the previous afternoon using a solar collector built from recycled PET bottles and aluminum cans — the same technology platform as the RTU (Universal Thermodynamic Revolution) project [12]. At dawn, they release the Pelican-UAV. The vehicle ascends silently, finds a thermal, and begins to soar. On the ground, a tablet displays the video stream and position data. The bird flies for hours. The animals below do not know they are being watched. The vessels offshore do not know they have been documented.

When the system detects an anomaly — smoke, machinery movement, vehicular intrusion, a vessel running dark — it transmits a geo-referenced alert. The ranger has evidence, has coordinates, has a timestamp. With that they can act, file a report, and build a legal record.

That is the actual design. Everything else — the composite materials, the VBC, the structural battery — is engineering in service of that moment.

9. Applications Matrix

Sector	Specific Application
Ecosystem Conservation	Detection of illegal logging, agricultural encroachment and poaching. Monitoring of biological corridors, threatened wildlife populations and vegetation health (NDVI). Operated by community rangers without specialized infrastructure.
Marine Protected Areas / IUU	Persistent surveillance of fishing exclusion zones. Detection of vessels operating without valid AIS transponder. Evidence collection for maritime authority reporting. Zero radar detectability at operational altitude.
Wildlife Research	Zero-bias behavioral observation: the platform is indistinguishable from local avifauna. Applicable to migration, feeding, reproduction and disturbance-response studies.
Wildfire Early Warning	Continuous hotspot detection in remote areas. Real-time coordinate transmission for rapid brigade response.
Atmospheric Boundary	In-situ measurement of pressure, temperature and humidity gradients

Layer	in the ABL over remote ecosystems.
Emergency Communications	Low-latency communications relay in rural or disaster-affected areas, leveraging high endurance as a low-altitude platform.

10. Limitations and Future Work

This document constitutes a conceptual pre-design technical specification. The following limitations must be explicitly acknowledged:

- Absence of physical prototype: performance values (L/D, MTOW, VBC efficiency, RCS estimates) are design estimates without experimental validation.
- Aerodynamic simplification: the analysis does not incorporate CFD simulations; behavior under turbulent regime, boundary layer separation and wingtip effects require detailed numerical modeling.
- H₂ risk management: hydrogen as lift gas implies safety protocols (leak detection, certified envelope materials) not developed in this version.
- Structural battery: solid polymer electrolyte integrated in composite laminate presents TRL 3–4 maturity; production-scale viability requires independent validation.
- RCS estimates: radar cross-section values are qualitative engineering estimates based on material and geometric analogy with avian targets; rigorous quantification requires anechoic chamber measurement of a physical prototype.

Future development lines: (1) CFD simulation of the wing profile at low Re regime; (2) lab-scale structural laminated battery cell prototyping; (3) VBC subsystem bench testing with a selected micro-compressor; (4) H₂ permeation barrier material analysis compatible with solid polymer electrolyte; (5) RCS measurement of scale prototype in controlled RF environment.

11. Conclusion

Project Pelican-UAV redefines the concept of aerial persistence through multidisciplinary integration. By unifying energy storage in the structural skin of the vehicle, delegating lift to a static H₂ volume, proposing a closed-loop pneumatic altitude control system, and adopting a flight behavior indistinguishable from the pelagic avifauna of the operational environment, the platform simultaneously solves three problems that conventional UAVs address separately and inadequately: endurance, ecological non-interference, and low observability.

The result is a silent, persistent, open-source observation vector — charged from solar infrastructure built from recycled urban waste — that can be operated by a community ranger in a remote ecosystem with no specialized training, and that an IUU fishing vessel's radar system will identify as a bird.

Future work is oriented toward progressive independent validation of each subsystem following an incremental Technology Readiness Level (TRL) methodology consistent with the resources of an open, distributed research program.

Data Availability

This preprint contains no experimental data or primary datasets. The complete document, including previous versions and supplementary materials, will be available under CC BY-SA 4.0 license at the Zenodo repository upon DOI assignment. Source code for future simulations will be published at the Fundación Hijos del Sol y la Basura GitHub repository (hdbasura25-oss.github.io/Hijosdelabasura).

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